Akshay Narayan

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EDUCATION

Massachusetts Institute of Technology. *2016 — 2022*
M.S. Computer Science, 2019
Ph.D. Computer Science, 2022
Thesis Title: *Enabling Configurable, Extensible, and Modular Network Stacks*
Advisors: Hari Balakrishnan, Mohammad Alizadeh.

University of California, Berkeley. *2011 — 2015*
Bachelor of Science in Electrical Engineering and Computer Science with High Honors, 2015.

PROFESSIONAL APPOINTMENTS

Brown University, Providence, RI, USA
Assistant Professor, Computer Science. *July 2024 —*
University of California, Berkeley, Berkeley, CA, USA
Postdoctoral Research Fellow, NetSys Lab. *August 2022 — June 2024*
Staff Researcher, NetSys Lab, advised by Sylvia Ratnasamy and Scott Shenker. *Spring 2014 — August 2016*
Microsoft, Redmond, WA, USA
Research Intern in Mobility and Networking, advised by Behnaz Arzani. *May 2018 — August 2018*
Software Development Engineer Intern. Operating Systems Group. *May 2014 — August 2014*
Jive Software, Palo Alto, CA, USA
Engineering Intern. *May 2013 — August 2013*
NASA Ames Research Center, Mountain View, CA, USA
Education Associates Program. *May 2012 — November 2012*

HONORS AND AWARDS

Google	ML and Systems Junior Faculty Award	*2026*
ACM EuroSys	Best Artifact Award	*2021*
ACM SIGCOMM	Best Student Paper Award	*2018*
National Science Foundation	NSF Graduate Research Fellowship	*2017*
Massachusetts Institute of Technology	Jacobs Presidential Fellowship	*2016*
University of California, Berkeley	Boeing Scholar	*2014*
University of California, Berkeley	Eta Kappa Nu	*2012*
University of California, Berkeley	Regents and Chancellors Scholar	*2011*

PUBLICATIONS

Conferences are the primary publication venue in Computer Science. By convention, author order lists students and post-docs first (ordered by contribution), followed by faculty in no fixed order. Some publications list faculty authors in alphabetical order or order of contribution, others list the lead faculty author last.

- Corinn Tiffany, Artem Agvanian, Ziyun Song, Isabella Szabo, Kinan Dak Albab, Will Crichton, Philip Levis, **Akshay Narayan**, Deepti Raghavan, Malte Schwarzkopf
  "We Need to Talk About the Actual Guarantees of Rust-based Systems"
  PLOS 2026

- Franco Solleza, Justus Adam, **Akshay Narayan**, Malte Schwarzkopf, Andrew Crotty, Nesime Tatbul
  "Kernel Extension DSLs Should Be Verifier-Safe!"
  eBPF 2025

- Rahul Bothra, Venkat Arun, Brighten Godfrey, **Akshay Narayan**, Ahmed Saeed
  "Lightweight Automated Reasoning for Network Architectures"
  HotNets 2024

- Margarida Ferreira, Ranysha Ware, Yash Kothari, Inês Lynce, Ruben Martins, **Akshay Narayan**, Justine Sherry
  "Reverse-Engineering Congestion Control Algorithm Behavior"
  IMC 2024

- Lloyd Brown, Albert Gran Alcoz, Frank Cangialosi, **Akshay Narayan**, Mohammad Alizadeh, Hari Balakrishnan, Eric Friedman, Ethan Katz-Bassett, Arvind Krishnamurthy, Michael Schapira, Scott Shenker
  "Principles for Internet Congestion Management"
  SIGCOMM 2024

- Lloyd Brown, Yash Kothari, **Akshay Narayan**, Arvind Krishnamurthy, Aurojit Panda, Justine Sherry, Scott Shenker
  "How I Learned to Stop Worrying About CCA Contention"
  Hotnets 2023

- William Sussman, Emily Marx, Venkat Arun, **Akshay Narayan**, Mohammad Alizadeh, Hari Balakrishnan, Aurojit Panda, Scott Shenker
  "The Case for an Internet Primitive for Fault Localization"
  Hotnets 2022

- Prateesh Goyal, **Akshay Narayan**, Frank Cangialosi, Srinivas Narayana, Mohammad Alizadeh, Hari Balakrishnan
  "Elasticity Detection: A Building Block for Internet Congestion Control"
  SIGCOMM 2022

- Margarida Ferreira, **Akshay Narayan**, Inês Lynce, Ruben Martins, Justine Sherry
  "Counterfeiting Congestion Control Algorithms"
  Hotnets 2021

- Frank Cangialosi, **Akshay Narayan**, Prateesh Goyal, Radhika Mittal, Mohammad Alizadeh, Hari Balakrishnan
  "Site-to-Site Internet Traffic Control"
  EuroSys 2021

- **Akshay Narayan**, Aurojit Panda, Mohammad Alizadeh, Hari Balakrishnan, Arvind Krishnamurthy, Scott Shenker
  "Bertha: Tunneling through the Network API"
  Hotnets 2020

- Hongzi Mao, Parimarjan Negi, **Akshay Narayan**, Hanrui Wang, Jiacheng Yang, Haonan Wang, Ryan Marcus, Ravichandra Addanki, Mehrdad Khani Shirkoohi, Songtao He, Vikram Nathan, Frank Cangialosi, Shaileshh Venkatakrishnan, Wei-Hung Weng, Song Han, Tim Kraska, Mohammad Alizadeh
  "Park: An Open Platform for Learning-Augmented Computer Systems"
  NeurIPS 2019

- **Akshay Narayan**, Frank Cangialosi, Deepti Raghavan, Prateesh Goyal, Srinivas Narayana, Radhika Mittal, Mohammad Alizadeh, Hari Balakrishnan
  "Restructuring Endpoint Congestion Control"
  SIGCOMM 2018

- Saksham Agarwal, Shijin Rajakrishnan, **Akshay Narayan**, Rachit Agarwal, David Shmoys, Amin Vahdat
  "Sincronia: Near-Optimal Network Design for Coflows"
  SIGCOMM 2018

- **Akshay Narayan**, Frank Cangialosi, Prateesh Goyal, Srinivas Narayana, Mohammad Alizadeh, Hari Balakrishnan
  "The Case for Moving Congestion Control Out of the Datapath"
  HotNets 2017

- Peter Gao, **Akshay Narayan**, Sagar Karandikar, Joao Carreira, Sangjin Han, Rachit Agarwal, Sylvia Ratnasamy, Scott Shenker
  "Network Requirements for Resource Disaggregation"
  OSDI 2016

- Peter Gao, **Akshay Narayan**, Gautam Kumar, Rachit Agarwal, Sylvia Ratnasamy, Scott Shenker
  "pHost: Distributed Near-Optimal Datacenter Transport Over Commodity Network Fabric."
  CoNEXT 2015

TEACHING

Below, "Instructor effectiveness rating" is the percentage of students who answered either "Agree" or "Strongly Agree" to the statement "Overall, I rate this instructor as effective." in an end-of-semester survey.

Designing High-Performance Network Systems (CSCI 1675), Brown University. *Spring 2025, Fall 2025*

- Newly-developed advanced systems course.
- Spring 2025: 21 students; Instructor effectiveness rating: 92%

Computer Networks and the Internet (CSCI 2680), Brown University. *Spring 2026, Fall 2024*

- Newly-developed graduate seminar course.
- Spring 2026: 19 students; Instructor effectiveness rating: 100%
- Fall 2025: 5 students; Instructor effectiveness rating: 100%

ADVISING

Ph.D. research advisor, Brown University:

Zixuan Peng	*September 2026*	—	
Oğuzhan Çölkesen	*September 2025*	—	
Co-Advised with Nikos Vasilakis.			
Yuchen Lu	*May 2025*	—	
Comprehensive Exam Committee Member. Co-Advised with Deepti Raghavan and Nikos Vasilakis.			
Justus Adam	*May 2026*	—	
Thesis Committee Member.			
Morgan Borjigin-Wang	*September 2025*	—	
Comprehensive Exam Committee Member. Co-Advised with Deepti Raghavan and Malte Schwarzkopf.			
Franco Solleza	*February 2025*	—	*August 2026*
Thesis Committee Member. Co-Advised with Malte Schwarzkopf.			

M.Sc. research advisor, Brown University:

Ziyun "Alice" Song → UPenn (PhD)	*October 2024*	—	*August 2026*

Undergraduate research advisor, Brown University:

Bhavani Venkatesan	*May 2025*	—	
Alexander Khosrowshahi	*October 2025*	—	
Edward Nguyen	*February 2025*	—	
Mengistie Hailemariam	*November 2024*	—	
Hee Su "Julie" Chung → Google Honor's Thesis Advisor.	*September 2025*	—	*May 2026*
Katie Li Honor's Thesis Reader.	*April 2026*	—	*May 2026*
Roberto Gonzales Matos → Adobe Honor's Thesis Advisor.	*September 2025*	—	*May 2026*
Megan Zheng → Princeton (PhD)	*January 2025*	—	*May 2026*

Research advisor, Massachusetts Institute of Technology:

Aditi Srinivasan → Jane Street Capital Undergraduate and M.Eng. student Thesis: Measuring and Optimizing for Network Conditions on Drones	*October 2018*	—	*December 2020*

PROFESSIONAL SERVICE

University and Departmental Service

- Brown University, Advanced Undergraduate Research Fellowships Review Committee: 2025
- Brown University, Dept. of Computer Science, UTRA Faculty Mentor: Summer 2025, Fall 2025, Summer 2026
- Brown University, Dept. of Computer Science, Brown Systems Week Faculty Mentor: Summer 2025, Summer 2026
- Brown University, Dept. of Computer Science, PhD Admissions Committee: 2024, 2025
- Brown University, Dept. of Computer Science, exploreCSR Faculty Mentor: Fall 2024

Conferences are the primary publication venue in Computer Science. Service to the profession centers on conference program committee memberships.

Conference and Workshop Program Committees

- NINeS: 2027
- HotNets: 2026, 2024
- eBPF: 2026, 2025
- EuroSys: 2027
- APNet: 2026, 2024
- SIGCOMM: 2026, 2025
- CoNEXT: 2026, 2025
- NSDI: 2026, 2024
- SysDW: 2024
- ANRW: 2024, 2023

Journal Reviewer

- SIGCOMM CCR: 2025, 2024

Artifact Evaluation Committee

- SIGCOMM: 2021

NSF Panelist

- CISE: 2026, 2025